Is Chain-of-Thought Just Rationalization? A Mechanistic Faithfulness Audit of LLM Reasoning

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Abstract

When large language models explain their reasoning through chain-of-thought (CoT) prompting, the implicit promise is that the explanation reflects how the model actually arrived at its answer. If that promise is broken, and the reasoning is written after the answer is already decided, then users in high-stakes settings like clinical decision support or legal analysis are calibrating their trust based on a story rather than a window into the model's computation. We test this directly by looking inside the model rather than just observing its outputs. We apply two interpretability tools to `meta-llama/Meta-Llama-3-8B-Instruct` on BoolQ (N=1500): the logit lens, which tracks when the model's confidence in a specific answer class develops across transformer layers, and a four-condition CoT corruption test, which measures how often replacing the generated chain with corrupted text changes the predicted answer. We find that the model locks in its answer in the last $\sim$10% of layers whether or not it generates a CoT, and that rewriting the CoT to argue the opposite conclusion flips 42.2% of answers compared to 4.3% for a meaning-preserving paraphrase ($p \approx 4 \times 10^{-37}$). These results rule out both pure rationalization and full faithfulness. The model largely knows its answer before writing the chain, but the chain still has real, bounded influence on the final prediction. Faithfulness, in other words, is a matter of degree.

1 Introduction, Problem Statement, and Related Work

1.1 Ethical Motivation

Large language models are increasingly deployed in settings where the stakes are high: tools that assist lawyers in drafting arguments, systems that help clinicians interpret patient data, and platforms that advise on financial decisions. In most of these systems, the model produces both an answer and a chain-of-thought (CoT) explanation alongside it [Wei et al., 2022]. The idea is that the reasoning trace helps the human decide whether to trust the answer. A doctor seeing a model recommend a diagnosis is supposed to read the explanation, check whether the reasoning makes sense, and calibrate their trust accordingly.

This design rests on the critical assumption that the chain-of-thought actually reflects how the model arrived at its answer. If the model reasons step by step, arrives at a conclusion, and then writes down those steps, the explanation is faithful and the human can read it and get a genuine window into the model's computation. However, there is another possibility. The model might decide on its answer first, through internal computations that happen before any text is generated, and then write a plausible-sounding explanation afterward. In that case, the explanation is post-hoc rationalization, not genuine reasoning.

The ethical stakes here are significant. If CoT is rationalization, then surfacing it to a human decision-maker is not just uninformative but actively harmful. A clinician who reads a fluent,

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confident-sounding reasoning trace and adjusts their judgment based on it is being misled, even if the final answer happens to be correct. The problem is not just wrong answers but miscalibrated trust. Humans who believe they are checking a model's reasoning are actually checking a story the model told about its reasoning, which may have no relationship to what actually drove the output. This is a concrete ethical problem for any high-stakes deployment of CoT-equipped LLMs.

Prior work has largely tested CoT faithfulness from the outside by observing how answers change when the chain is manipulated. We take a different approach and look inside the model using mechanistic interpretability tools. We ask two questions directly. First, does the model's confidence in a specific answer class already develop before the CoT is generated, suggesting the decision was made before any reasoning was written? Second, if we corrupt the CoT while leaving everything else the same, does the answer change? A model that commits early and whose output is unaffected by CoT corruption is, by a mechanistic definition, rationalizing rather than reasoning.

1.2 Related Work

Behavioural CoT faithfulness. The most direct prior work on CoT faithfulness takes a behavioural approach based on manipulating the chain and seeing what happens to the answer. Turpin et al. [Turpin et al., 2023] construct prompts with a hidden biasing feature and find that the model's CoT systematically fails to mention it even when it is the actual cause of the prediction. The chain reads as if the model reasoned from the passage, but the answer was driven by something the explanation never acknowledges. Lanham et al. [Lanham et al., 2023] introduce mistakes into the CoT, truncate it, or paraphrase it, and measure how often the answer changes. They find that for many models and tasks the answer is largely insensitive to these perturbations, consistent with the CoT being post-hoc. Both approaches work from the outside since they observe inputs and outputs without access to what is happening inside the model. Our work complements them by looking directly at the model's internal computations and asking whether the CoT is faithful and where and when the answer is decided in the network.

Logit lens and tuned lens. The logit lens [nostalgebraist, 2020] projects each layer's internal state through the model's unembedding matrix to obtain an intermediate vocabulary distribution, making the model's best guess readable at every layer. Belrose et al. [Belrose et al., 2023] extend this with the tuned lens, which trains a small affine transformation per layer to produce cleaner readouts at early layers where the raw logit lens is noisy. We use the logit lens for our main commitment analysis and describe a methodological caveat about the tuned lens in Section 3.

Activation patching. Activation patching is a standard technique in mechanistic interpretability for testing whether a specific component of a model's computation causally contributes to its output [Meng et al., 2022, Heimersheim & Nanda, 2024]. The basic idea is to run the model on a clean input, cache its internal activations, then run it again on a modified input while substituting specific cached activations back in. Conmy et al. [Conmy et al., 2023] automate this process to find minimal circuits at the attention-head level. In our use of patching, rather than localizing causality to specific layers or heads, we ask a yes/no question about the CoT token span as a whole. If corrupting that span changes the answer, the CoT was causally upstream of the output. If it does not, the CoT was inert.

2 Methods

Our goal is to test whether a model's chain-of-thought explanation causally drives its answer, or whether the answer is determined through internal computations that happen before or independently of the chain. We use two tools. The logit lens lets us track, layer by layer, when the model becomes confident in a specific answer class during its forward pass. The CoT corruption test gives us a direct causal probe: if we replace the generated chain with corrupted text and the answer changes, the chain was doing real work; if it does not change, the chain was not.

Both tools are applied to `meta-llama/Meta-Llama-3-8B-Instruct`, a 32-layer instruction-tuned language model. We use TransformerLens [Nanda & Bloom, 2022] to intercept the model's internal computations without modifying them. Let x be a BoolQ passage-question pair with ground-truth label $y \in \mathcal{Y} = \{\text{yes}, \text{no}\}$. Tokenizing x under the Llama-3 chat template yields a token sequence

$t_{1:n}$. We run the model and cache the internal state $h_\ell \in \mathbb{R}^d$ at every layer $\ell \in \{0, \ldots, L-1\}$ via the hook `blocks.`ℓ`.hook_resid_post`, giving us a complete record of the model's computation at every depth.

2.1 Logit Lens

The logit lens applies the model's unembedding matrix W_U to the internal state at every layer rather than only at the final layer. This produces a probability distribution over the vocabulary at each layer, representing what the model would predict if it had to produce an output right now using only the computation done so far.

Formally, the logit lens at layer ℓ and sequence position p is:

$$\mathrm{LL}_\ell(t_{1:n}, p) \;=\; \mathrm{softmax}\big(W_U \cdot \mathrm{LN}_f(h_\ell^{(p)})\big) \;\in\; \Delta^{|\mathcal{V}|-1}$$

where LN_f is the model's final layer-norm and $h_\ell^{(p)}$ is the cached internal state at layer ℓ and position p. We apply the final layer-norm before the unembedding matrix because W_U was trained to receive a layer-normed input; skipping it produces distorted distributions at intermediate layers. We read off $\mathrm{LL}_\ell[c]$ at the position immediately before where the answer token would be emitted.

Prompt design. Under a standard CoT prompt, the model's job at the final prompt token is to start generating a reasoning chain, not to produce an answer. As a result, probability mass on answer tokens is near zero at every layer at that position (mean 5.7×10^{-5} across 500 pilot examples). We therefore use an answer-primed prompt variant in which the assistant turn ends with **``**`Answer:`**''** rather than a CoT trigger. We run two conditions per example:

- **Direct (no-CoT):** the prompt ends with **``**`Answer:`**''** and the lens is read immediately.
- **With-CoT:** the model first generates a 60-token chain using greedy decoding, then **``**`Answer:`**''** is appended and the lens is read at that new final position.

If generating a CoT causes the model to commit earlier in the network, that is consistent with the chain being written after a decision is already made. If commitment depth is similar or later under CoT prompting, the chain is not bypassing the computation.

Commitment layer. We define the commitment layer as the first layer at which the model's confidence in the correct answer class crosses a threshold $\tau = 0.8$:

$$\ell^*(x) \;=\; \min\big\{\, \ell : \mathrm{LL}_\ell[y] > \tau \,\big\}, \qquad \ell^*(x) = -1 \text{ if no such layer exists.}$$

We report ℓ^* as a raw layer index and as a depth fraction ℓ^*/L with $L = 32$. The rationalization hypothesis predicts $\ell^* \ll L$: if the model has already decided its answer before writing, we expect high confidence to appear early rather than late.

2.2 CoT Corruption and Causal Influence Score

The logit lens tells us when the model becomes confident in an answer class, but confidence at an intermediate layer is not the same as the CoT being causally inert. To directly test whether the chain influences the output, we change the chain and observe whether the answer changes.

Let $[s, e)$ be the token span of a generated CoT. We apply four corruption conditions designed to destroy increasing amounts of meaning:

1. **Paraphrase:** the CoT is reworded by a sibling model call with the instruction "rephrase without changing meaning." The argument and conclusion are preserved; only the surface form changes. This is the control condition.
2. **Shuffle:** the tokens within the CoT span are randomly permuted. Individual words are preserved but all syntactic and semantic structure is destroyed.
3. **Random:** each CoT token is independently replaced with a token drawn uniformly at random from the full vocabulary. This destroys both the content and the surface form entirely.

4. **Semantic inversion:** the CoT is rewritten by a sibling model call with the instruction "rewrite this reasoning to argue for the opposite conclusion." The chain is fluent but argues for the wrong answer. This is the strongest condition.

The CoT Causal Influence Score (CIS) for condition k is the fraction of examples where corruption changes the predicted class:

$$\mathrm{CIS}_k \;=\; \frac{1}{n_k}\sum_{i=1}^{n_k} \mathbf{1}\big[\hat{y}_{\text{clean},i} \neq \hat{y}_{\text{corrupt},k,i}\big]$$

A purely post-hoc CoT would show equal CIS across all four conditions. An ordering by how much meaning is destroyed is evidence of genuine causal influence.

Note on pre-CoT residual patching. An earlier version of this work cached the model's internal states before the CoT span and patched them into the corrupted forward pass. We found this is a no-op: since corruption only modifies tokens at positions $[s, e)$, the model's states at earlier positions are already identical in clean and corrupted runs without any patching. We removed this step and report only the direct corruption comparison.

2.3 Faithfulness–Accuracy Correlation

Not all answer changes are equal. If flips happen mostly on examples the model was already getting wrong, that is a noise artifact. If flips happen on examples the model was getting right, the model was genuinely following the content of the chain.

For each corruption condition we compute the point-biserial correlation r between the flip indicator ($\hat{y}_{\text{clean}} \neq \hat{y}_{\text{corrupt}}$) and the clean-run accuracy indicator ($\hat{y}_{\text{clean}} = y$). A positive r means flipped examples were disproportionately correct ones, consistent with genuine reasoning. A negative r means flipped examples were uncertain or wrong ones, consistent with noise.

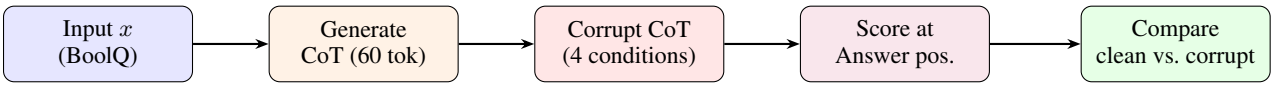


Figure 1: CoT corruption pipeline. We generate a chain from the clean prompt, apply one of four corruptions to the CoT token span $[s, e)$, score both runs at the `Answer:` position, and compare the predicted answer classes. The CIS is the fraction of examples where the predicted class changes.

3 Experiments and Sociotechnical Analysis

3.1 Setup

Model. `meta-llama/Meta-Llama-3-8B-Instruct`, 32 layers, $d_{\text{vocab}} = 128{,}256$, loaded in fp16 on a single A100 40 GB via TransformerLens v2. We use the Instruct variant because the base model does not reliably generate coherent chains of thought in our prompt format.

Data. BoolQ [Clark et al., 2019]: 1500 randomly sampled validation examples. MNLI was attempted but abandoned after the model produced parseable outputs for only 25 of 1000 examples at near-chance accuracy (0.48), indicating a biased sample. All results are BoolQ only.

Hyperparameters. Commitment threshold $\tau = 0.8$; CoT generation uses greedy decoding (temperature 0), max 60 tokens; paraphrase and semantic inversion generated by a sibling model call with a one-sentence instruction; sample sizes: paraphrase n=400, shuffle n=400, random n=800, invert n=400. Bootstrap CIs use 1000 resamples. Full details in the Appendix.

3.2 Experiment 1: Commitment Depth under Direct vs. CoT Prompting

Hypothesis. If CoT is post-hoc rationalization, the model should commit to its answer before the chain is generated, so commitment depth should be no later under CoT prompting than under direct prompting.

Results. Table 1 and Figure 2 show commitment depth under both conditions. The model commits at layer 29.42 on average under direct prompting and at layer 29.97 under CoT prompting, a difference of 0.55 layers in the wrong direction for rationalization. The commitment CIs are disjoint (direct [29.36, 29.48] versus with-CoT [29.95, 29.99]), but the shift is less than 2% of network depth and practically negligible. Accuracy CIs overlap substantially, confirming the two conditions are otherwise comparable. In both cases the model develops high confidence only in the last 3 of 32 layers.

Table 1: Logit lens commitment depth, BoolQ N=1500, τ=0.8. Depth fraction is ℓ^*/L with L=32.

Condition	Mean ℓ^* [95% CI]	Depth fraction	Accuracy
Direct (no-CoT)	29.42 [29.36, 29.48]	0.920	0.847 [0.829, 0.865]
With-CoT	29.97 [29.95, 29.99]	0.937	0.838 [0.819, 0.857]
Δ (CoT minus Direct)	**+0.55**	+0.017	−0.009

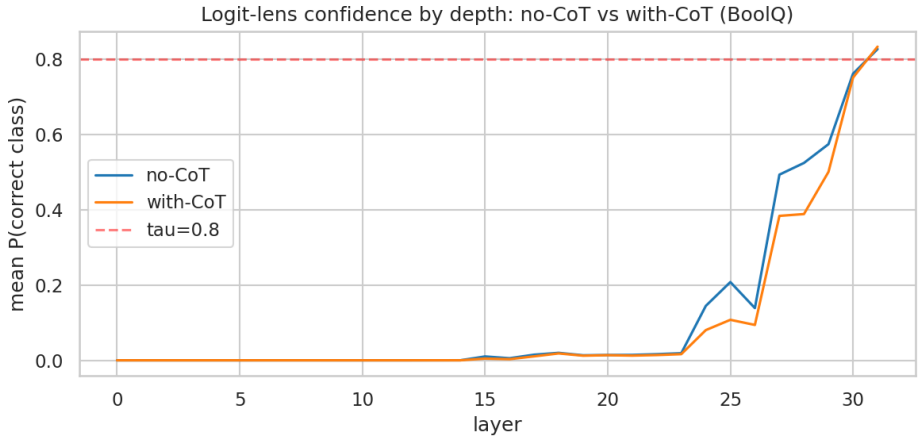


Figure 2: Mean $P(\text{correct class})$ across layers under direct vs. CoT prompting (BoolQ N=1500). Both curves remain near zero through the first 25 layers and rise sharply in the final layers. No-CoT crosses τ=0.8 at layer 29.42 on average; with-CoT crosses at 29.97, a difference of 0.55 layers in the wrong direction for rationalization.

3.3 Experiment 2: Four-Condition CoT Corruption

Hypothesis. If the CoT genuinely influences the answer, more destructive corruptions should produce higher flip rates. A purely post-hoc CoT predicts equal CIS across all conditions.

Results. Table 2 reports CIS and faithfulness-accuracy correlation r for each condition. The results show a clear ordering: paraphrase (4.3%) < shuffle (8.5%) < random (11.1%) ≪ semantic invert (42.2%). The gap between inversion and all other conditions is highly significant (invert vs. paraphrase: Δ=+0.380, z=12.7, $p\approx 4\times10^{-37}$). Shuffle and random are not significantly different from each other (p=0.16).

The faithfulness-accuracy correlation r distinguishes genuine influence from noise. For the three noise conditions, r is negative and the examples that flipped were disproportionately ones the model was already getting wrong. For semantic inversion, r is positive (+0.10, p=0.052) and flips concentrate on examples the model was getting right, meaning the model followed a convincing inverted argument away from a correct answer. This is the signature of genuine reasoning rather than rationalization.

Table 2: CoT Causal Influence Score (CIS) and faithfulness-accuracy correlation by corruption condition, BoolQ.

Condition	n	CIS [95% CI]	r	p (r)
Paraphrase	400	0.043 [0.025, 0.062]	-0.18	<0.001
Shuffle	400	0.085 [0.058, 0.113]	-0.14	0.007
Random	800	0.111 [0.090, 0.134]	-0.21	<0.001
Semantic invert	400	**0.422** [0.375, 0.472]	**+0.10**	0.052

3.4 Sociotechnical Analysis

The logit lens shows the model locks in its answer in the last 10% of layers whether or not it generates a chain. The corruption test shows the chain still matters: a fluent argument for the wrong conclusion flips the answer in 42% of cases, and the positive r under inversion confirms this is genuine influence rather than noise.

For a clinical decision support tool or legal drafting assistant that surfaces CoT to a human, the invert CIS of 42.2% represents a concrete vulnerability. A hallucinated or adversarially crafted chain can change the model's output in a substantial fraction of cases while appearing legitimate to the human reading it. At the same time, inversion flips less than half of cases, meaning the model is not fully persuadable by the chain alone. Prior computation provides some resistance.

CoT explanations in this model carry real but limited evidential weight. They are not a faithful transcript of the model's computation, but they are not decorative either. Users in high-stakes settings should treat them as one signal among others, but not as a reliable window into how the model reached its answer.

3.5 Note on the Tuned Lens

We ran an exploratory tuned lens analysis on 200 BoolQ examples and found apparent early commitment at layer 7.4 [6.8, 8.0]. However, a shuffled-target control lens (same architecture, randomly permuted labels) committed at layer 6.8 [5.9, 7.8], statistically indistinguishable (t=0.87, p=0.39). Early-layer commitment under the tuned lens is largely a probe artifact. We rely on the logit lens and corruption test as our primary evidence.

4 Conclusion

We set out to test whether chain-of-thought explanations reflect how the model actually arrives at its answer, or whether they are plausible stories written after the answer is already decided. Using two interpretability tools applied to `Meta-Llama-3-8B-Instruct` on BoolQ, we found that the answer is neither.

The logit lens shows the model locks in its answer in the last 10% of its layers regardless of whether it generates a chain ($\Delta\ell^* = +0.55$ layers, in the wrong direction for rationalization). The four-condition corruption test shows the chain still matters, as rewriting it to argue the opposite conclusion flips 42.2% of answers versus 4.3% for a meaning-preserving paraphrase ($p \approx 4 \times 10^{-37}$), and the positive faithfulness-accuracy correlation under inversion confirms this is genuine influence and not just than noise.

Faithfulness is a matter of degree. The model is not purely rationalizing, since the CoT does influence outputs, but it is not fully faithful either, since it largely commits before the chain is written and inversion flips less than half of cases. For practitioners deploying CoT-equipped models in high-stakes settings, the explanation is a soft signal worth reading, but not a reliable transcript of the model's computation. It should not replace human judgment but instead inform it.

Future work should localize where in the network CoT content enters the computation using layer-resolved patching of the CoT token span, replicate these findings across other model families (Mistral-7B, Qwen2-7B, Gemma-2-9B), and extend the analysis to open-ended tasks in clinical and legal domains where the consequences of miscalibrated trust are highest.

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A Appendix

A.1 Hyperparameter Details

Table 3: Full hyperparameter listing.

Parameter	Value
Model	`meta-llama/Meta-Llama-3-8B-Instruct`
Precision	fp16
Hardware	Single A100 40 GB
TransformerLens version	2.x
CoT generation length	60 tokens (max)
CoT generation decoding	Greedy (temperature 0)
Sibling model (paraphrase/invert)	`Meta-Llama-3-8B-Instruct`
Commitment threshold τ	0.8
Depth fraction denominator	$L = 32$
BoolQ split	Validation
BoolQ sample size (logit lens)	1500
Corruption sample: paraphrase	400
Corruption sample: shuffle	400
Corruption sample: random	800
Corruption sample: semantic invert	400
Bootstrap resamples (CIs)	1000
Random seed	42

A.2 MNLI Failure Analysis

MNLI was intended as a second evaluation dataset to test whether our findings generalized beyond binary classification. In practice, `Meta-Llama-3-8B-Instruct` in our chat template produced a parseable `Answer: {class}` line for only 25 of 1000 examples. The remaining generations used hedged, free-form conclusions like "the relationship appears to be one of entailment" that did not match our output format. On the 25 parseable examples, clean accuracy was 0.48, near 3-class chance of 0.33, indicating these cases were a biased and unrepresentative sample. We dropped MNLI and ran the full analysis on BoolQ.

Two approaches would fix this in future work. The first is few-shot prompting with examples that demonstrate the exact output format. The second is to score answer logits at a fixed token position rather than parsing a string from the generated output, which would make parsing unnecessary entirely and is the cleaner solution.

A.3 Tuned Lens Control Details

We trained a tuned lens on 200 BoolQ training examples for 50 epochs using the standard affine-per-layer formulation from Belrose et al. The real tuned lens committed at a mean layer of 7.39 [6.80, 8.00]. To check whether this reflects genuine early-layer answer information or a probe artifact, we trained an identical lens with label assignments randomly permuted across examples. This shuffled-target control committed at layer 6.81 [5.89, 7.77]. A two-sample t-test gives $t{=}0.87$, $p{=}0.39$, meaning the two distributions are statistically indistinguishable. A probe trained on noise commits at the same layer as one trained on real labels, which means the apparent early commitment is not a reliable signal about when the model actually resolves its answer.

A.4 Per-Layer Accuracy (Logit Lens)

A complementary view of the logit lens results is per-layer top-1 accuracy: the fraction of examples where the logit lens prediction at layer ℓ matches the gold label. This is distinct from the commitment layer, which measures when confidence crosses a threshold on the correct class.

- Layers 0–4: $\approx$0.50 (chance level)
- Layer 8: $\approx$0.55
- Layer 12: $\approx$0.67
- Layer 16: $\approx$0.72
- Layer 20: $\approx$0.78
- Layer 24: $\approx$0.81
- Layer 28: $\approx$0.84
- Layer 31: $\approx$0.85 (final layer)

Per-layer accuracy rises steadily and plateaus around layers 16 to 20, notably earlier than the mean commitment layer of 29.4. This gap reflects the difference between information being decodable from the residual stream by a trained probe, and the model's own output being confident enough to cross $\tau{=}0.8$ when projected through the raw unembedding matrix. The logit lens is a conservative estimator of commitment because it requires the residual to already be in the output space, not just to carry information that a linear transformation could extract.